DEEPTHIKA S

Portfolio

Email: deepthikas123@gmail.com
LinkedIn: deepthika365
Google Scholar | GitHub: deepthikas123-ops

Research Profile

Passionate interdisciplinary researcher working at the intersection of **design**, **technology** and **human behaviour**. My work examines how people perceive, interact with and respond to intelligent systems, spanning **human-AI interaction**, **LLM behaviour**, **generative AI**, **immersive technologies** and **computational design**. Through computational methods and experimental studies, I aim to understand **human perception**, **preferences and decision-making**, bridging technological research with human-centred applications.

Education

Amrita Vishwa Vidyapeetham, Coimbatore — Aug 2023 - June 2027 (expected)
B.Tech Electronics and Communication Engineering — *CGPA: 7.99/10 (through Semester 6)*

Vidhya Niketan Public School — Jan 2022 - Mar 2023
CBSE Class XII (Physics, Chemistry, Biology, Mathematics) — *93.4%*

SSVM World School — Mar 2020 - Apr 2021
CBSE Class X — *97.4%*

Research Experience

Design Research Intern – SCALE Lab, CPDM, Indian Institute of Science (IISc) — Jan 2026 - June 2026

- Conceptualized the **Atomic User Model (AUM)**, a structured representation of a person using a core-identity nucleus and four interpretable layers to support personality-aware interaction with LLMs.
- Developed the **Iconicity Classification Algorithm (IKA)**, a computational framework for characterising product designs along a spectrum from conventional to unconventional using image embeddings and vision-language models.
- Designed and conducted a **cross-generational study** investigating neophilic and neophobic preferences towards conventional and unconventional product aesthetics across Generations X, Y, Z, and Alpha.
- Contributed to **FLAVOVR**, an immersive VR eye-tracking system for capturing implicit aesthetic preferences and exploring how individual gaze behaviour can inform personalised product design, alongside a separate study of aesthetic gaze behaviour.

Selected Research and Manuscripts

KAAYA: The Body Draws What the Eye Never Sees — **Accepted**
NeurIPS 2026, Creative AI Track (Agency), Sydney

- Developed a wearable IMU-to-generative-art system that splits drawing's sensorimotor loop between a blindfolded artist and an AI, using hand kinematics to create traces that a generative model visually interprets, and examines the resulting shift in creative agency.

Where the Gaze Lingers: From Immersive Free-Viewing to a Stable Aesthetic Fingerprint — **Accepted**
VRST 2026 (Sendai, Japan), Poster and Demo Track

- Built FLAVR, an immersive gaze-contingent VR system, and showed that the non-salient residue of gaze identifies individuals about three times better than saliency-driven gaze; combining gaze residuals with VLM-derived aesthetic signatures further improves identification.

Neophilic and Neophobic Design Preferences Across Generational Cohorts: Consumer Attraction to Conventional and Unconventional Product Aesthetics — **Submitted**
Journal of Business Research, Elsevier

- Developed a computational framework combining immersive gaze-based preference capture and vision-language models to study how consumers respond to conventional versus unconventional product aesthetics across generations.

The Calculator Fallacy: Learners Blindly Trust AI, Even When It Is Wrong — **Submitted**
16th International Conference of the European Academy of Design (TRUST___DISTRUST)

- Studied how learners calibrate their trust in conversational AI when it provides information with varying levels of accuracy, examining misinformation absorption, confidence, reliance, and checking behaviour.

When the Majority Is Wrong: Would AI Still Follow the Crowd? Studying Conformity and Authority Bias in Multi-Agent LLM Systems — **Submitted**
13th IEEE UPCON 2026, IIT Kanpur

- Adapted the Asch conformity paradigm to multi-agent LLM systems to study whether AI agents maintain their own judgement when surrounded by a unanimous group, and how perceived authority influences their decisions.

Creating an Atomic User Model for Personality-Aware LLM Interaction

- Proposed a personality-aware user model for LLM interaction that represents a person's stable identity through interpretable psychological, cognitive, behavioural, and social layers, enabling more personalised retrieval and generation.

VAPOURS: Sculpting Fog to Find the Form Before the Form

- Proposed a VR medium for early-stage product form exploration using dissipating volumetric fog, where maintaining a form over time becomes a way to capture the designer's attention and intent.

Design and Engineering Projects

Tactile Trails: Sensory Playmat for Neurodivergent Children (Cognify-ReWire, IIT Madras)

- Designed a cognitive-sensory playmat with Velcro-detachable zones on a bamboo base, refined with neurologists, child-development experts, and toy design mentors; ran structured playtests with children and parents. Showcased at Shaastra 2025 and selected as a Top 25 finalist out of 1200+ teams.

SETU 5.0: A Rapid Deployment System for Emergency Response

- Engineered a robotic cable system for last-mile delivery of medical supplies in disaster zones, as an all-weather alternative to aerial drones, with a web-based teleoperation interface and ESP32 firmware for motor control, obstacle detection, and payload verification. **Tech:** ESP32, PlatformIO, load cell, ultrasonic sensor

Multi-Level Morse Code Logic Game

- Designed a multi-level digital access system using Morse-code inputs, DIP switches, and XNOR/AND logic, with progressive challenges from 2-bit decoding to an 8-bit dual-input authentication scheme.

ECOSHIELD

- Designed a biomimetic UV-filtering concept inspired by edelweiss fibres and moth-eye nanostructures, using a layered 3D model to explore UV filtering while maintaining visible-light transparency.

Skills Summary

Programming & Development:	Python, MATLAB/Simulink, JavaScript/TypeScript (React, Next.js), HTML, CSS, LaTeX, Supabase, Git/GitHub
Machine Learning & AI:	Computer vision and transfer learning (MobileNetV3, TinyML), embedding-based similarity analysis, LLM APIs (OpenAI, Gemini), prompt engineering, multi-agent LLM simulation, generative AI, LLM evaluation, dataset annotation (Roboflow)
HCI, XR & Embedded:	VR with eye-tracking (Meta Quest Pro), gaze-contingent interaction, human-AI interaction, haptic interface design, ESP32, Arduino, IMU sensing, sensor integration, haptic actuation, agent-based simulation (NetLogo)
Design & Product:	Figma, Fusion 360, Canva, UI/UX design, product prototyping, assistive and sensory product design, biomimetic design, design research, ergonomic design, graphic design

Honors and Awards

- **Cognify (ReWire), Shaastra 2025, IIT Madras – Top 25 Finalist** out of 1200+ teams (Tactile Trails).
- **Pitch Arena – Third Place**, Anokha 2026, for pitching "Tactile Trails" (organised by ELITE, the Entrepreneurship Club of Amrita Vishwa Vidyapeetham).
- **Anokha 2026 Techfair:** "SETU 5.0" shortlisted for the Open House display, with positive feedback from CEOs and NITI Aayog officials.
- **Prompt Masters '25 – Second Runner Up:** built HelloMahila, an AI-driven women's wellness companion using structured prompt flows.

Certifications

- **Build Wireframes and Low Fidelity Prototypes**, *Google UX Design Professional* — June 2025
- **Fundamentals of Graphic Design**, *California Institute of Arts* — June 2025
- **Introduction to the Internet of Things and Embedded Systems**, *University of California, Irvine* — May 2025
- **Innovation Through Design: Think-Make-Break-Repeat**, *University of Sydney* — May 2025
- **MATLAB Onramp Certification**, *MathWorks* — Dec 2024

Languages

- English *(Professional)*, Tamil *(Native)*, Hindi *(Limited)*, Italian *(Elementary)*, Japanese *(Elementary)*

Interests

- Reading, Junk Journaling, Sketching, Art & Calligraphy